Question 7: Configure continuous deployment using Kubernetes by integrating the deployment with the CI/CD pipeline and automating manifest application.

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-2

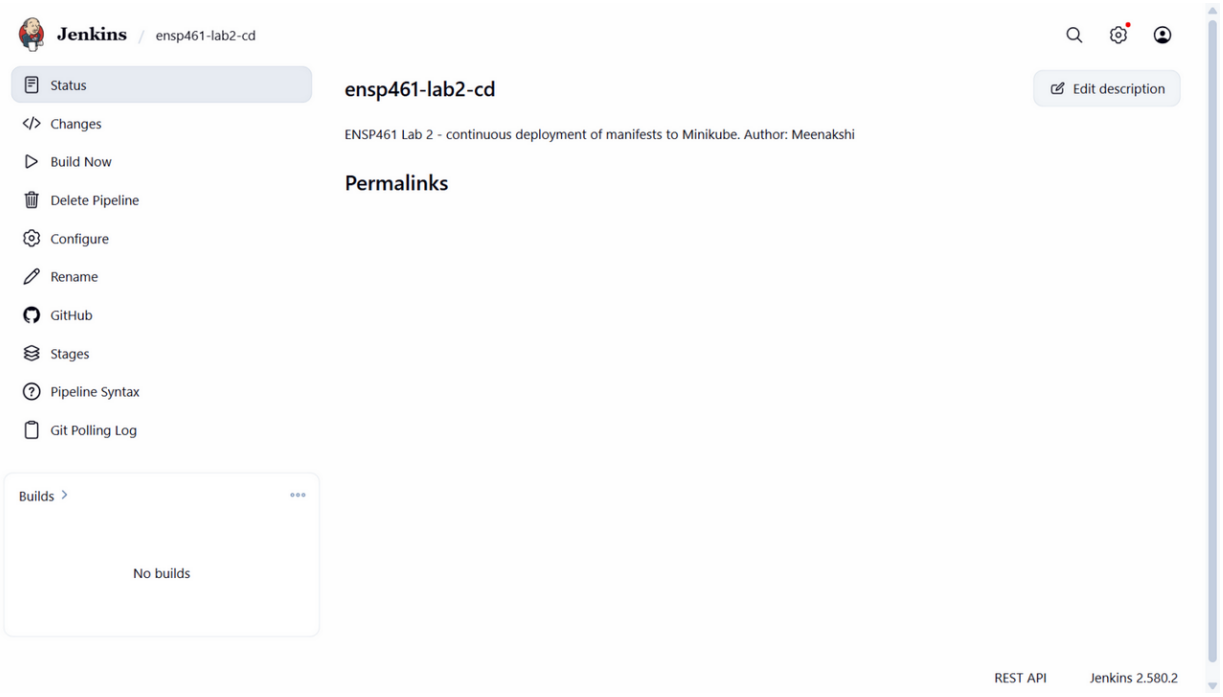
1. The CD pipeline (Jenkinsfile-CD)

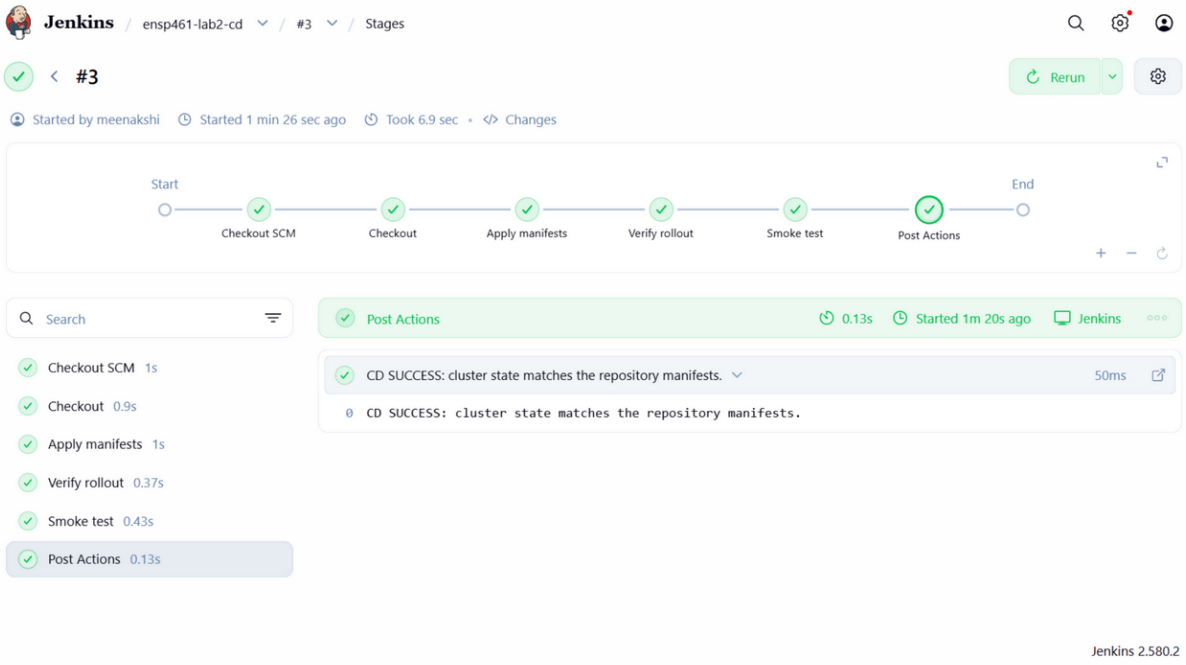
Jenkinsfile-CD (committed to the repository)

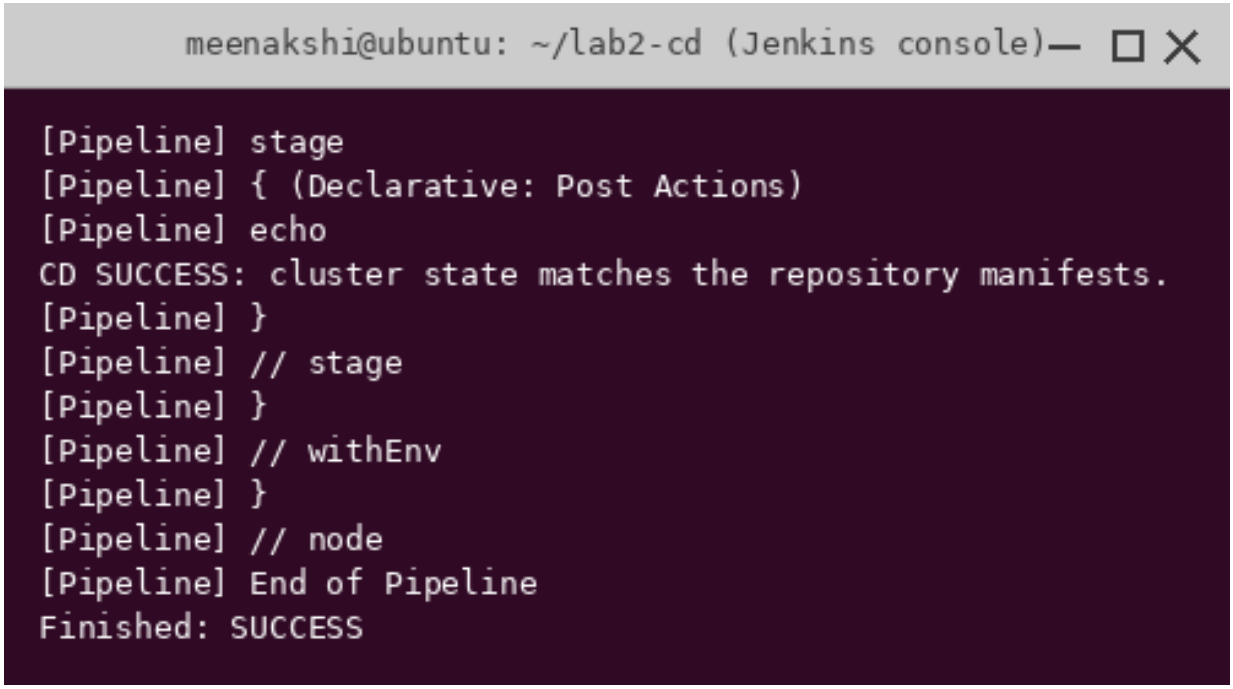
```
// ENSP461 Lab 2 - Continuous deployment to Kubernetes (Minikube)
pipeline {
    agent any
    stages {
        stage("Checkout") {
            steps {
                echo "Pulling manifests from GitHub..."
                checkout scm
            }
        }
        stage("Apply manifests") {
            steps {
                echo "Applying Kubernetes manifests to the Minikube cluster..."
                sh "kubectl apply -f k8s/configmap.yaml"
                sh "kubectl apply -f k8s/service.yaml"
                sh "kubectl apply -f k8s/deployment.yaml"
            }
        }
        stage("Verify rollout") {
            steps {
                sh "kubectl rollout status deployment/lab2-app --timeout=180s"
            }
        }
        stage("Smoke test") {
            steps {
                echo "Verifying the service responds..."
                sh "NODEIP=\$(kubectl get node minikube -o \
      jsonpath=\"{.status.addresses[?(@.type=='InternalIP')].address}\"); curl -sf \
      http://\$NODEIP:30080/health"
            }
        }
    }
    post {
        success { echo "CD SUCCESS: cluster state matches the repository manifests." }
        failure { echo "CD FAILED - check the stage logs." }
    }
}
```

- The Jenkins job ensp461-lab2-cd pulls this pipeline from the repository (Pipeline script from SCM) and polls GitHub every 5 minutes.
- The jenkins service account authenticates to the cluster with a kubeconfig installed at /var/lib/jenkins/.kube/config.

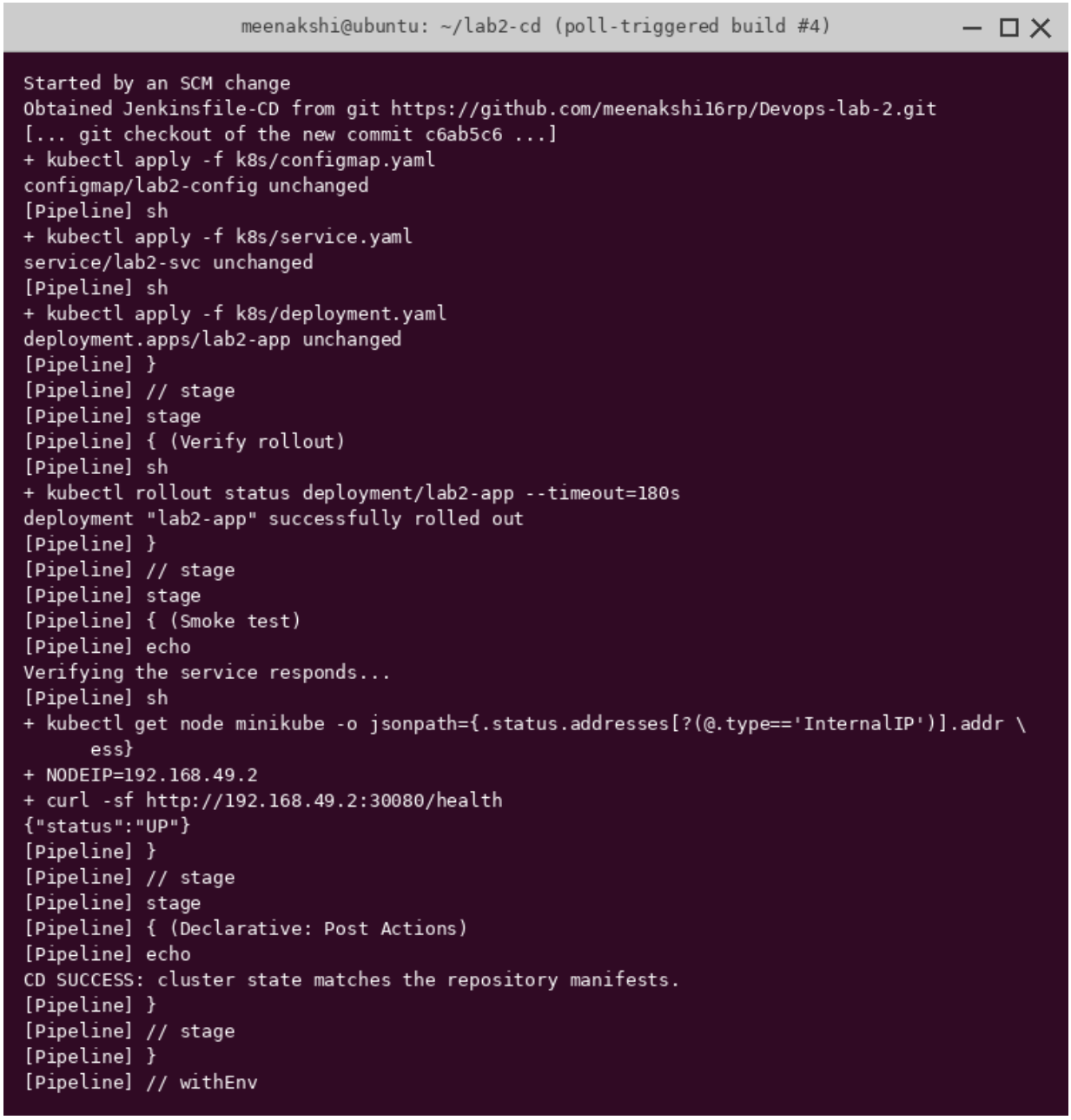
2. CD pipeline execution


Jenkins / ensp461-lab2-cd
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
GitHub
Stages
Pipeline Syntax
Git Polling Log
ensp461-lab2-cd
Edit description
ENSP461 Lab 2 - continuous deployment of manifests to Minikube. Author: Meenakshi
Permalinks
Builds
No builds
REST API
Jenkins 2.580.2


Jenkins / ensp461-lab2-cd / #3 / Stages
#3
Rerun
Started by meenakshi
Started 1 min 26 sec ago
Took 6.9 sec
Changes
Start
End
Checkout SCM
Checkout
Apply manifests
Verify rollout
Smoke test
Post Actions
Search
Checkout SCM 1s
Checkout 0.9s
Apply manifests 1s
Verify rollout 0.37s
Smoke test 0.43s
Post Actions 0.13s
Post Actions
0.13s
Started 1m 20s ago
Jenkins
CD SUCCESS: cluster state matches the repository manifests.
50ms
CD SUCCESS: cluster state matches the repository manifests.
Jenkins 2.580.2



3. Automated trigger from a git push



build #4 summary

```
Build #4 cause: Started by an SCM change
Result: SUCCESS (kubectl apply + rollout + smoke test)
```

4. Observations

- Build #1 failed in the smoke test: $(minikube ip) is unavailable to the jenkins user (profiles live in the user's home). Fix: resolve the node IP with kubectl.
- Build #2 failed again because the top-level .InternalIP field was removed in Kubernetes 1.37 - the jsonpath now reads .status.addresses[].
- Build #3 succeeded end to end; build #4 fired automatically from a git push (Started by an SCM change), completing the push-to-deploy loop.
- kubectl apply is idempotent: unchanged resources report unchanged, changed ones are updated in place.